

Laboratory of Applied Electronics: #1 Linear regulator

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Introduction

Modern designs require multiple power rails to supply electronic systems, with load requirements ranging from a few mA to over 100A. It is crucial to select the appropriate solution for the application, ensuring performance specifications like high efficiency, compact PCB layout, accurate output regulation, fast transient response, and low cost are met.

A power converter generates output voltage and current for the load from a given input power source. It needs to meet the load voltage or current regulation requirement during steady-state and transient conditions. It also must protect the load and system in case of a component failure.

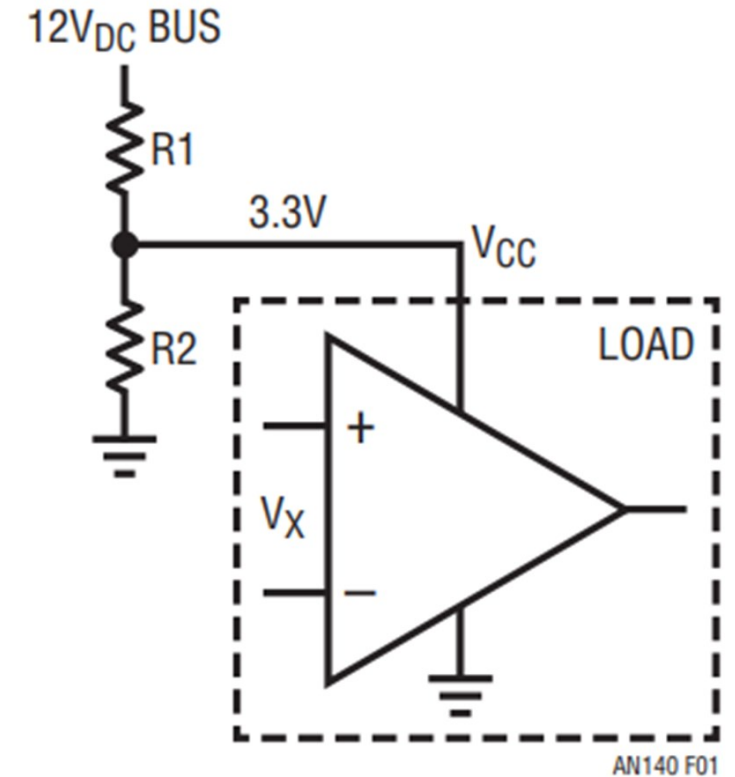
Depending on the specific application, a designer can choose either a linear regulator (LR) or a switching mode power supply (SMPS) solution. To make the best choice of a solution, it is essential to be familiar with the merits, drawbacks and design concerns of each approach.

Simple example

Consider an embedded system with a 12V bus rail available from the front-end power supply.

On the system board, a 3.3V output is needed to power an operational amplifier.

A simple approach is to use a resistor divider from the 12V bus; however, this method is often ineffective.



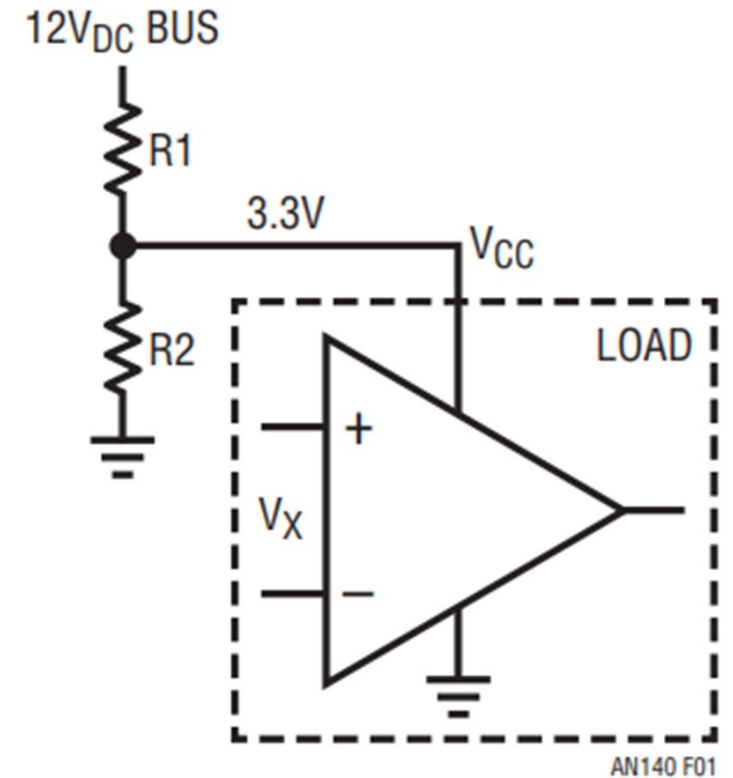
Simple example

The first problem is that the 12V bus input may not be well regulated (e.g., it may be the output of a rectifier).

There may be many other loads in the same system sharing the 12V rail. And the 12V bus voltage is sensitive to the bus loading conditions. As a result, a resistor divider cannot provide a regulated 3.3V to the op amp to ensure its proper operation.

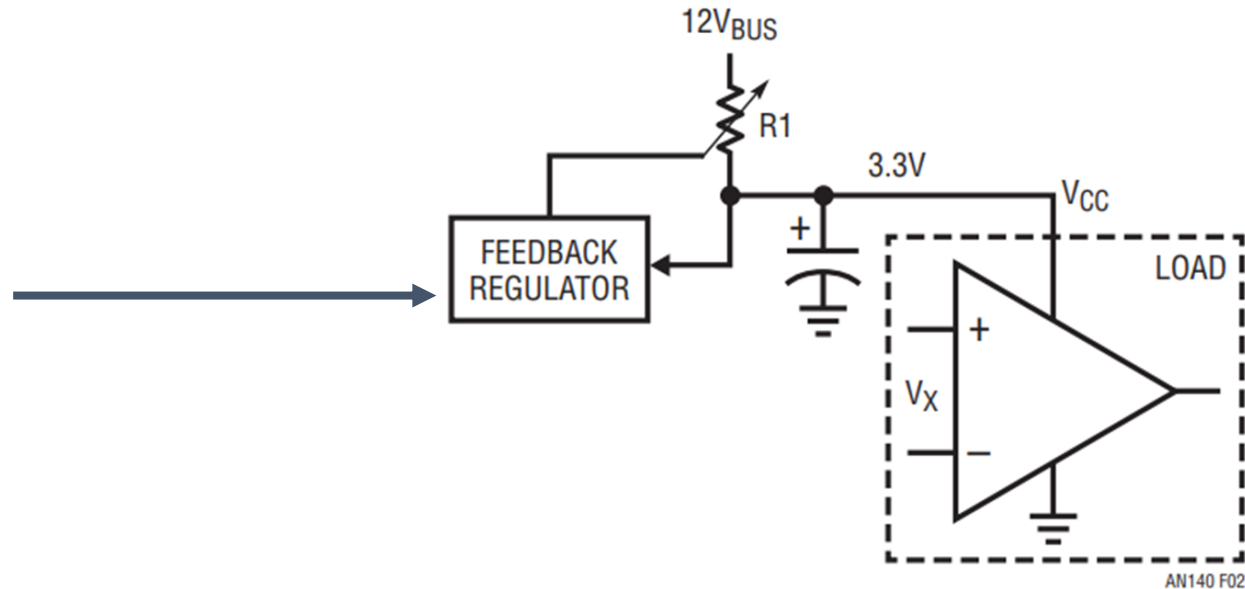
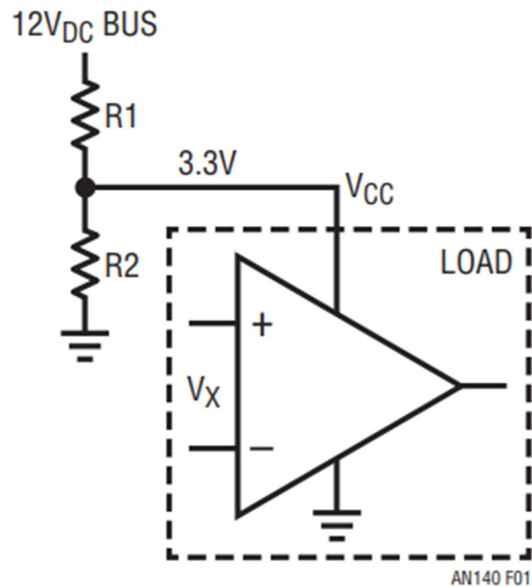
Moreover, the required current from the V_{CC} pin current may vary under different operating conditions, and voltage supplied by the divisor change with the load.

Therefore, a dedicated voltage regulation loop is needed. As shown in figure, the feedback loop needs to adjust the top resistor R_1 value to dynamically regulate the 3.3V on V_{CC} .



Simple example

As shown in figure, the feedback loop needs to adjust the top resistor R_1 value to dynamically regulate the 3.3V on V_{CC} .

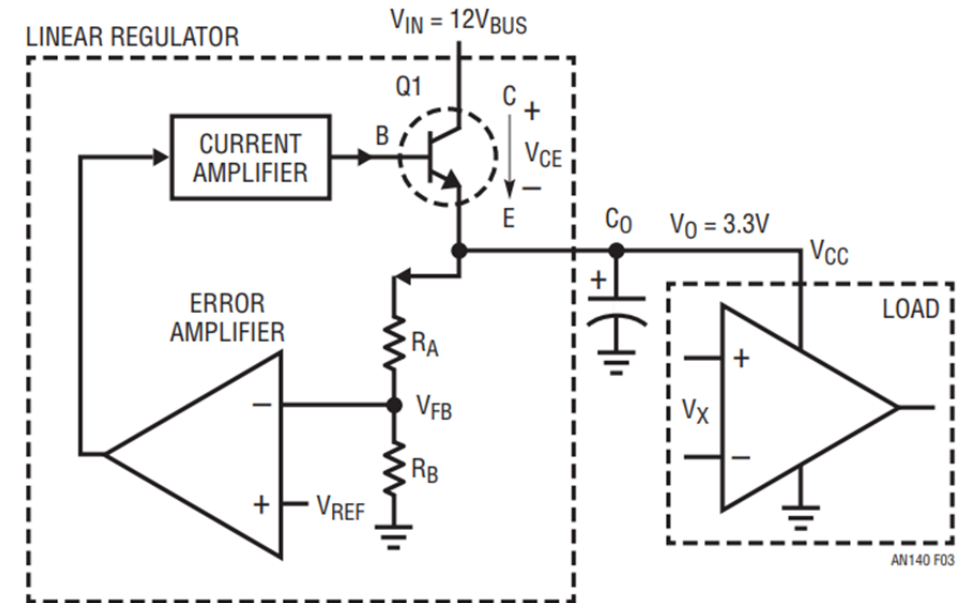


Linear regulator

A variable resistor can be implemented with a power transistor operating in linear mode, thus functioning as a variable resistor in series with the output load.

To create a feedback loop, an error amplifier senses the DC output voltage through a sampling resistor network and compares this feedback voltage to a reference voltage.

The error amplifier output voltage drives the base of the series power transistor via a current amplifier.



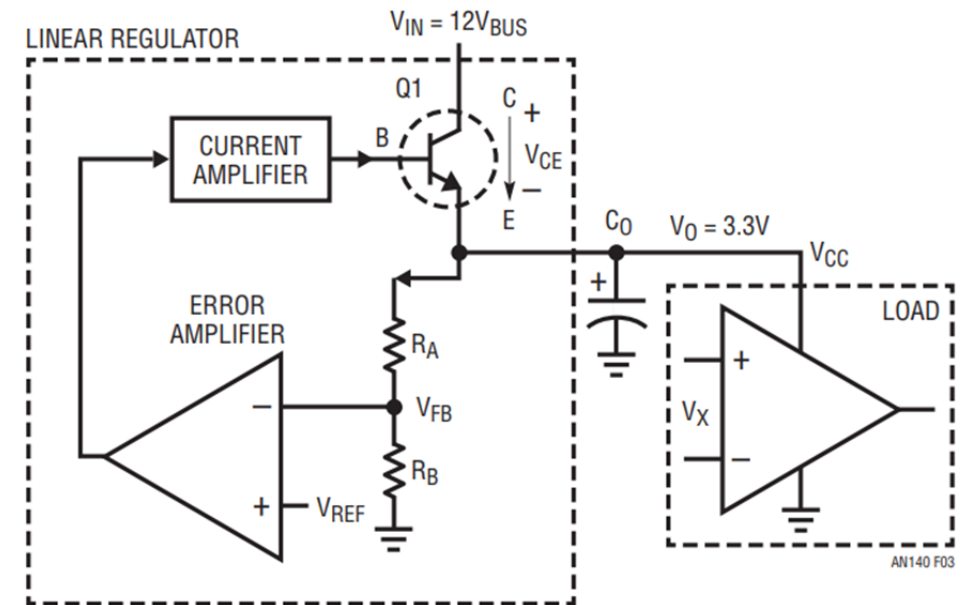
Linear regulator

When either the input voltage decreases or the load current increases, the output voltage goes down. The feedback voltage V_{FB} decreases as well.

As a result, the feedback error amplifier and current amplifier generate more current into the base of the transistor Q1. This reduces the voltage drop V_{CE} and hence brings back the V_{CC} output voltage, so that V_{FB} equals V_{REF} .

On the other hand, if the V_{CC} output voltage goes up, in a similar way, the negative feedback circuit increases V_{CE} to ensure the accurate regulation of the 3.3V output.

In summary, any variation of V_O is absorbed by the linear regulator transistor's V_{CE} voltage, and the output voltage is always constant.



Linear regulator

The linear regulator has been widely used for a very long time until switching mode power supplies became prevalent.

Even today, linear regulators are still widely used in a wide range of applications.

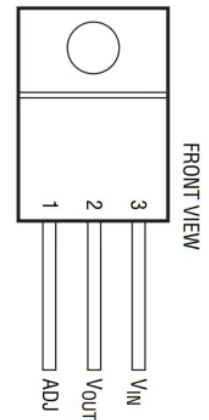
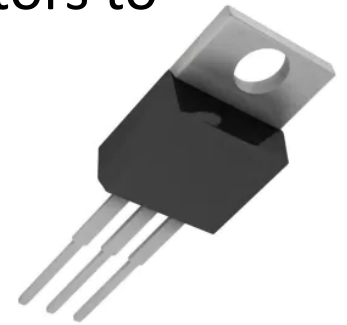
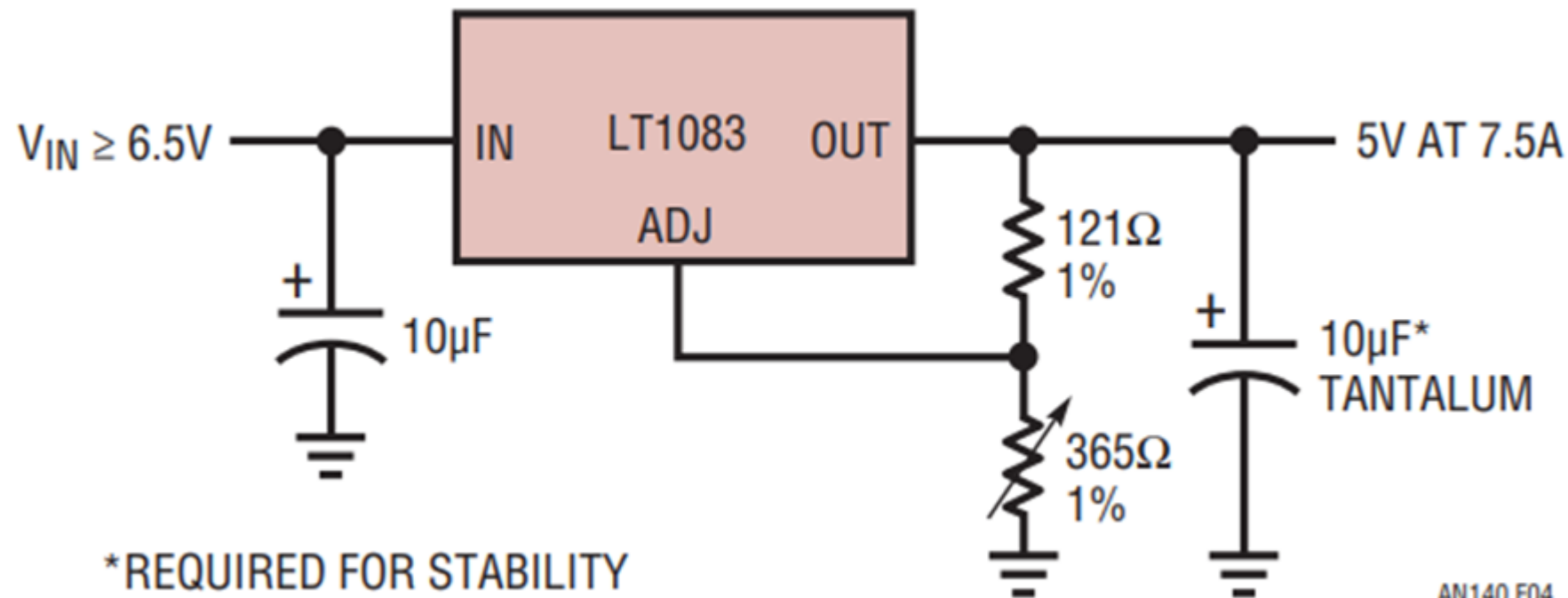
In addition to their simplicity of use, linear regulators have other performance advantages.

Power management suppliers have developed many integrated linear regulators. A typical integrated linear regulator needs only V_{IN} , V_{OUT} , an adjustment and optional GND pins.

Linear regulator

Figure shows a typical 3-pin linear regulator, the LT1083, which was developed more than 30 years ago.

It only needs an input capacitor, output capacitor and two feedback resistors to set the output voltage.



Limitations

A major drawback of using linear regulators can be the excessive power dissipation of its series transistor operating in a linear mode.

As explained previously, a linear regulator transistor is conceptually a variable resistor. Since all the load current must pass through the series transistor, its power dissipation is $P_{\text{LOSS}} = (V_{\text{IN}} - V_{\text{O}}) I_{\text{O}}$.

In this case, the efficiency of a linear regulator can be quickly estimated by:

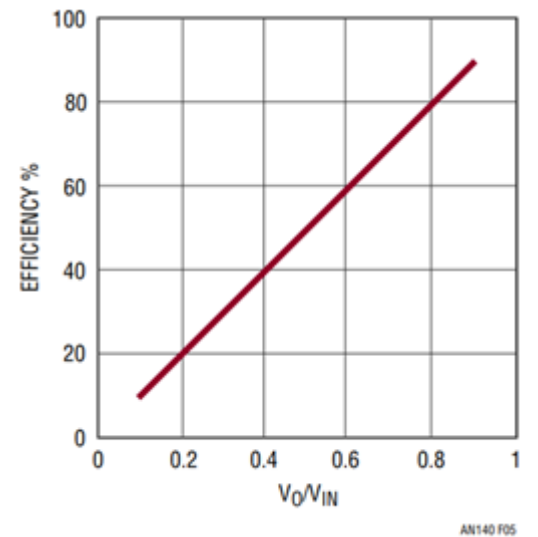
$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} = \frac{V_{\text{O}} I_{\text{O}}}{V_{\text{O}} I_{\text{O}} + (V_{\text{in}} - V_{\text{O}}) I_{\text{O}}} = \frac{V_{\text{O}}}{V_{\text{in}}}$$

Limitations

In this example, with a 12V input and a 3.3V output, the linear regulator's efficiency is only 27.5%. This means that 72.5% of the input power is dissipated as heat within the regulator.

Thus, the transistor must have the thermal capability to handle its power/heat dissipation at worst case at maximum V_{IN} and full load. So the size of the linear regulator and its heat sink may be large, especially when V_O is much less than V_{IN} .

Figure shows that the maximum efficiency of the linear regulator is proportional to the V_O/V_{IN} ratio.



Limitations

On the other hand, the linear regulator can be very efficient if V_O is close to V_{IN} .

However, the linear regulator (LR) has another limitation, which is the minimum voltage difference between V_{IN} and V_O .

The transistor in the LR must be operated in its linear mode. So it requires a certain minimum voltage drop across the collector to emitter of a bipolar transistor or drain to source of a FET. When V_O is too close to V_{IN} , the LR may be unable to regulate output voltage anymore.

Low dropout regulators

Linear regulators that can operate with low headroom ($V_{IN} - V_{OUT}$) are called low dropout regulators (LDOs).

It is important to note that linear regulators, including LDOs, can only provide step-down DC/DC conversion.

In applications that require V_O voltage to be higher than V_{IN} voltage or need negative V_O voltage from a positive V_{IN} voltage, linear regulators obviously do not work.

Use of linear regulator

There are applications in which linear regulators or LDOs provide superior solutions to switching supplies:

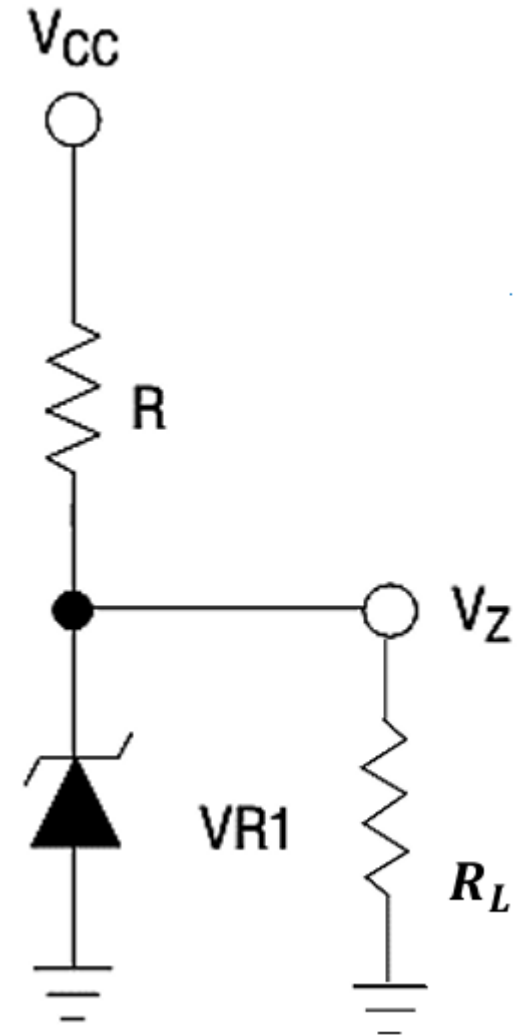
- **Simple/low cost solutions.** Linear regulator or LDO solutions are simple and easy to use, especially for low power applications with low output current where thermal stress is not critical.
- **Low noise/low ripple applications.** In noise-sensitive applications, such as communication and radio devices, minimizing supply noise is critical. Linear regulators produce very low output voltage ripple, as they lack frequently switching elements, and they can offer high bandwidth.
- **Low dropout applications.** For applications where output voltage is close to the input voltage, LDOs may be more efficient than an SMPS.

Zener diode as a stabilized power supply

The resistor R is connected in series with the Zener diode to limit current flow through the diode, with the voltage source connected across the combination.

The Zener diode is connected with its cathode terminal connected to the positive rail of the DC supply so it is reverse biased and will be operating in its breakdown condition. Resistor R is selected so to limit the maximum current flowing in the circuit.

With no load connected to the circuit, the load current will be zero, and all the circuit current passes through the Zener diode which in turn dissipates its maximum power.

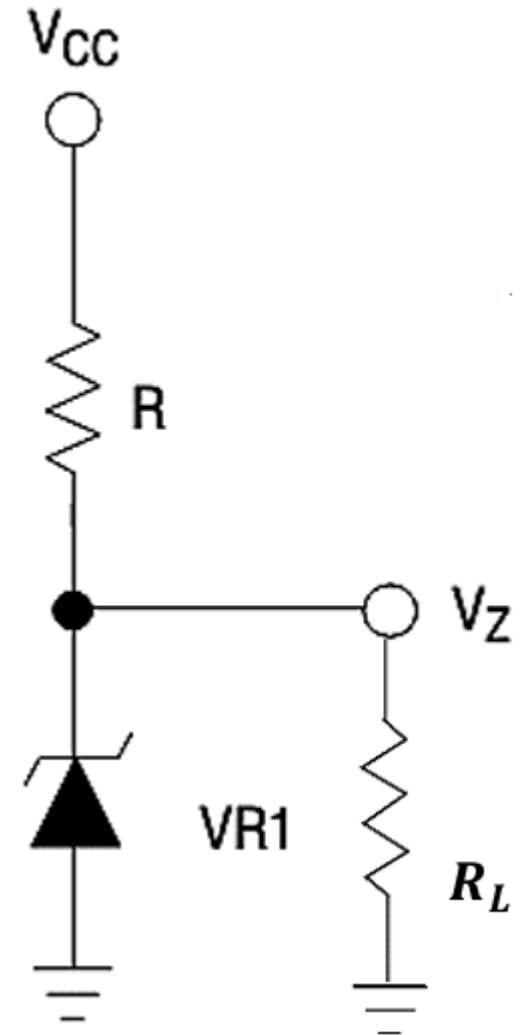


Zener diode as a stabilized power supply

Also a small value of the series resistor R will result in a greater diode current when the load resistance R_L is connected and large as this will increase the power dissipation requirement of the diode so care must be taken when selecting the appropriate value of series resistance so that the Zener's maximum power rating is not exceeded under this no-load or high-impedance condition.

The load is connected in parallel with the Zener diode, so the voltage across R_L is always the same as the Zener voltage.

There is a minimum Zener current for which the stabilization of the voltage is effective, and the Zener current must stay above this value always operating under load within its breakdown region. The upper limit depends upon the power rating of the device.

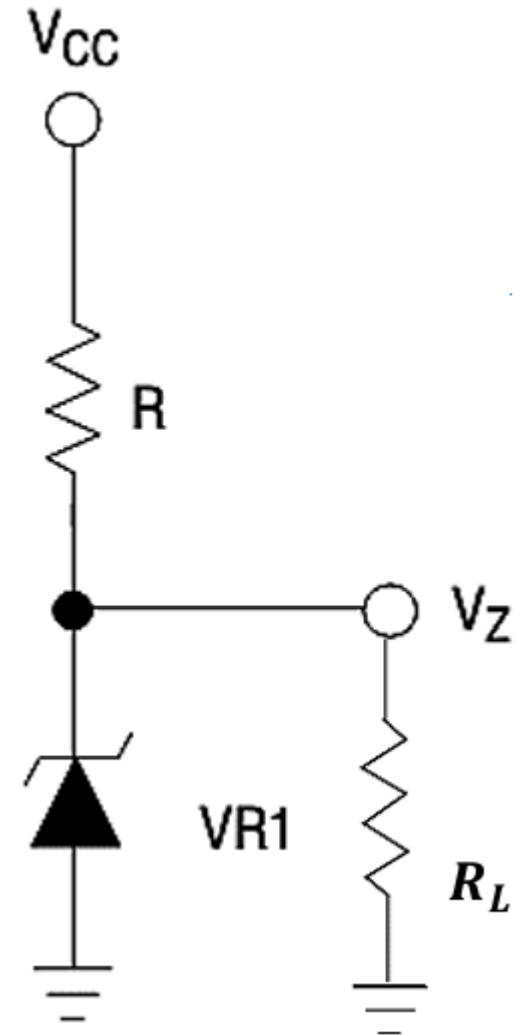


Zener diode as a stabilized power supply

A minor drawback of Zener diode stabilizer circuits is that the diode may generate electrical noise in the DC supply during voltage stabilization. Normally this is not a problem for most applications but the addition of a large value decoupling capacitor across the Zener's output may be required to give additional smoothing.

Another problem is the temperature dependency of the Zener's voltage.

The stabilized output voltage is always selected to be the same as the breakdown voltage V_Z of the diode



Single transistor series regulator

This circuit comprises a transistor and a Zener diode. In this setup, the load current flows through the series transistor.

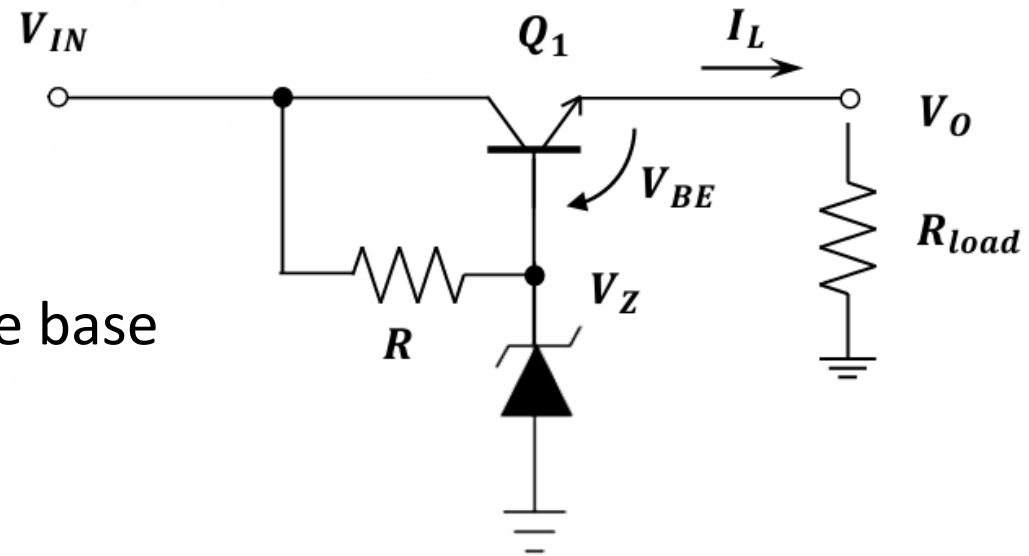
The series transistor acts as a variable resistor to achieve a regulated output voltage.

The Zener diode provides the reference voltage.

The operation this regulator relies on maintaining the base voltage at a stable level set by the diode.

The output voltage can be calculated as:

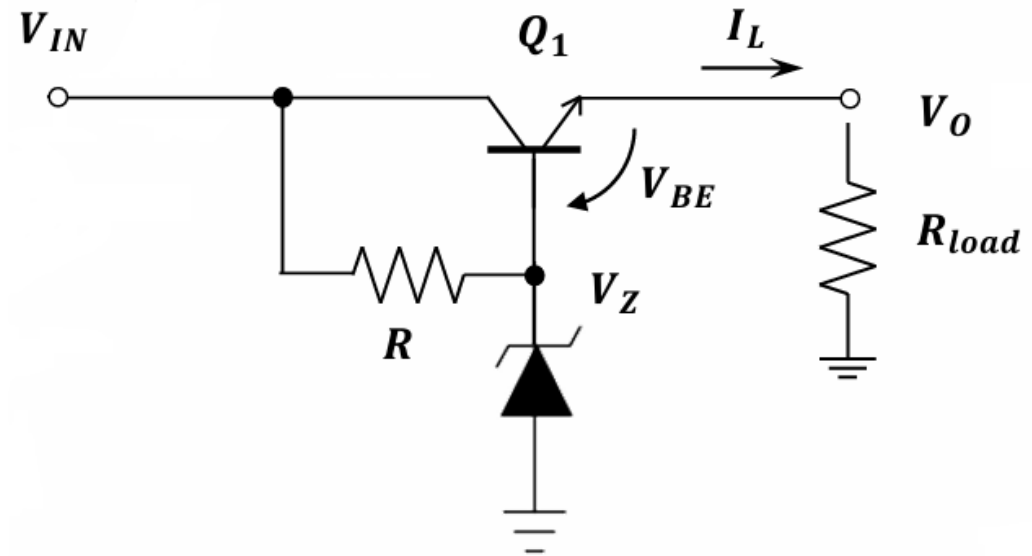
$$V_o = V_Z - V_{BE}$$



Single transistor series regulator

The output voltage regulation relies on the Zener voltage and the base-emitter (BE) junction forward drop.

This design can be improved by closing the feedback loop to enhance stability.

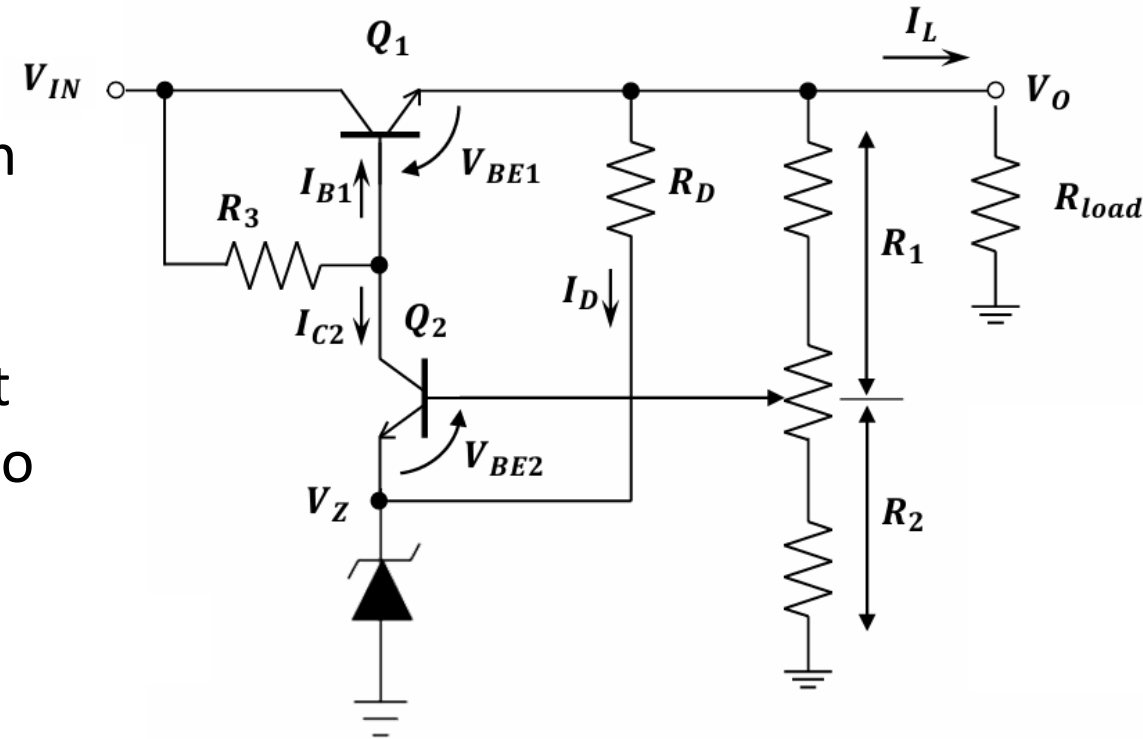


Two transistor closed-loop series regulator

To improve performance, a feedback network can be added to the voltage regulator circuit.

This is done by sampling the output, comparing it to a reference, and using a differential amplifier to feed back the difference and correct errors.

Although it is quite straightforward to use an operational amplifier, which will provide higher levels of feedback, and hence better regulation, but this two transistor circuit illustrates the principles well.

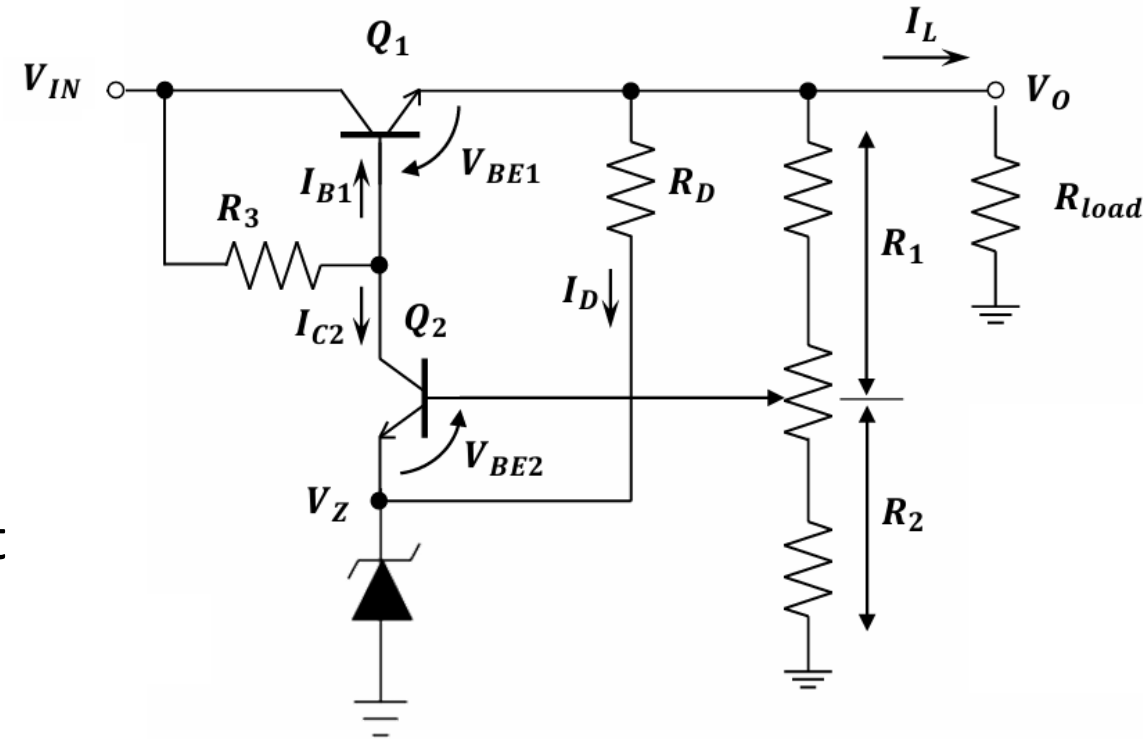


Two transistor closed-loop series regulator

Q1 forms the series pass transistor.

The second transistor, acts as the differential amplifier, feeding the error voltage between the reference diode and the sensed output voltage which is a proportion of the output voltage as set by the potentiometer.

The resistor, R_3 provides the current for the base of Q2 and R_D for the Zener diode.



Two transistor closed-loop series regulator

The output voltage in this circuit is not fixed and can be regulated through the output resistor divider.

$$V_{BE2} + V_Z = V_o \frac{R_2}{R_1 + R_2}$$

$$V_o = (V_{BE2} + V_Z) \left(1 + \frac{R_1}{R_2} \right)$$

